



Nguyen Hoang Viet

Viet Nguyen | hsgsstudent | Viet Nguyen | +84.971.161.827
 nguyenhoangviet.hsgs@gmail.com | hsgsstudent.github.io

SUMMARY

My research lies at the intersection of statistics and machine learning, where I seek to understand the behavior of machine learning models from a rigorous theoretical perspective. My recent work focuses on parameter convergence rates in Mixture-of-Experts models. I am also interested in quantum computing, especially quantum optimization and quantum machine learning. My long-term goal is to develop statistical methodologies that enhance the reliability and interpretability of machine learning systems across both classical and quantum computing platforms.

EDUCATION

2019 - 2022 **High School for Gifted Students, Hanoi University of Science (HUS)** Vietnam
High School Diploma (Specialized in Mathematics)
2022 - 2026 **Hanoi University of Science and Technology (HUST)** Hanoi, Vietnam
(Expected) BSc in Computer Science (Talented Program) (CPA: 3.89/4.0 - top 1% of the cohort)

RESEARCH EXPERIENCE

Independent Researcher Oct 2025 - present
– Supervisor: [Prof. Nhat Ho](#), The University of Texas at Austin
– Focused on the theoretical analysis of Mixture-of-Experts models, with particular emphasis on parameter convergence rates
– Conducted research on medical image retrieval methods, focusing on colorectal cancer detection
Undergraduate Research Assistant | NetLab - HUST Jul 2024 - Sep 2025
– Supervisor: [Prof. Trinh Chien](#), Hanoi University of Science and Technology
– Formulated the communication channel estimation problem using rigorous mathematical analysis and introduced an evolutionary algorithm to improve channel quality
– Conducted research using the D-Wave quantum computer to solve multi-objective optimization problems via quantum annealing

SELECTED JOURNAL PUBLICATIONS

[Advanced Quantum Annealing for the Bi-Objective Traveling Thief Problem: An \$\varepsilon\$ -Constraint-Based Approach](#)
Viet Nguyen, Nguyen Xuan Tung, Trinh Van Chien, Won-joo Hwang
IEEE Transactions on Quantum Engineering, 2026
[Improved Differential Evolution for Enhancing the Aggregated Channel Estimation of RIS-Aided Cell-Free Massive MIMO](#)
Trinh Van Chien, **Viet Nguyen**, Symeon Chatzinotas, Lajos Hanzo
IEEE Transactions on Vehicular Technology, 2025

Extending network lifespan and securing effective barrier coverage in 3D underwater wireless sensor networks
 Nguyen Thi My Binh, Le Duc Anh Vu, **Nguyen Hoang Viet**, Huynh Thi Thanh Binh, Ho Viet Duc Luong
Cluster Computing, 2026

SELECTED CONFERENCE PUBLICATIONS

A Mixture of Experts Vision Transformer for High-Fidelity Surface Code Decoding
Viet Nguyen, Manh Hung Nguyen, Hoang Ta, Van Khu Vu, Yeow Meng Chee
IEEE International Symposium on Information Theory (ISIT) 2026

PREPRINTS

Rethinking Multinomial Logistic Mixture of Experts with Sigmoid Gating Function
 Tuan Minh Pham*, Thinh Cao*, **Viet Nguyen***, Huy Nguyen, Nhat Ho**, Alessandro Rinaldo**
Under Review

A Statistical Theory of Gated Attention through the Lens of Hierarchical Mixture of Experts
Viet Nguyen*, Tuan Minh Pham*, Thinh Cao*, Huy Nguyen, Nhat Ho**, Alessandro Rinaldo**
Under Review

ACHIEVEMENT

2020	Vietnam Mathematical Olympiad	(Second Prize – ranked 13th)
2023	Vietnam Math Competition for University student	(First Prize – ranked 4th)
2025	Vietnam Math Competition for University student	(First Prize – ranked 8th)

LANGUAGES

Vietnamese	Native	
English	IELTS Overall 7.0	(L – R – W – S: 6.5 – 8.5 – 7.0 – 6.0)

REFERENCES

[Prof. Nhat Ho](#) The University of Texas at Austin
[Prof. Trinh Chien](#) Hanoi University of Science and Technology